

Spatial transcriptomics of kidney: a Xenium glomerulonephritis workflow

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One day face to face; the notebooks are an end-to-end reference you keep working through.

- We analyse one real Xenium kidney slide: 480-gene panel, FFPE, 8 tissue samples from three glomerulonephritis (GN) types.
- Six notebooks: read and split the slide, transcriptomics, classification, integration, differential expression, spatial analysis.
- Each notebook carries method notes and interpretation; solutions are revealed later.

Why spatial, and why these samples

Dissociated scRNA-seq tells you *which* cells are present; the kidney's function and its diseases live in *where* they are: glomeruli, tubules, and immune foci.

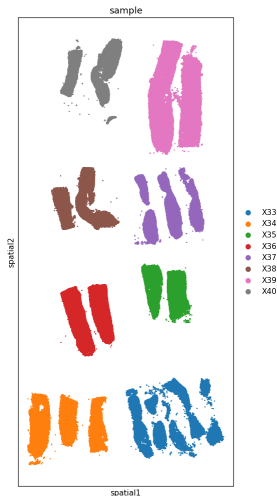
The slide carries 8 cores from three immune-mediated GN types:

- **ANCA-GN** (necrotising/crescentic), 5 samples
- **anti-GBM** (GBM), 1 sample
- **Lupus nephritis** (SLE), 2 samples

No healthy control is on this slide, so we contrast *ANCA-GN vs the other GN types*.

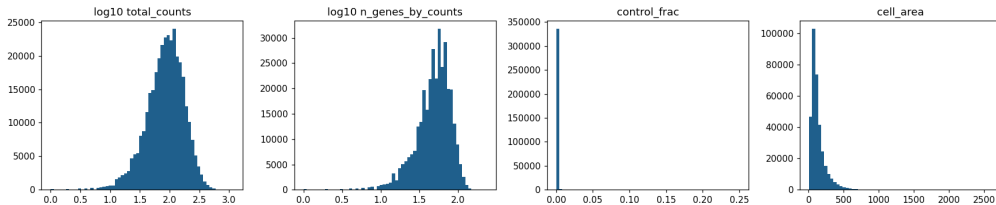
Xenium in one line. An imaging assay that localises individual transcripts of a targeted panel and assigns them to cells by segmentation: single-cell resolution, limited gene set, with coordinates kept.

One slide, eight samples



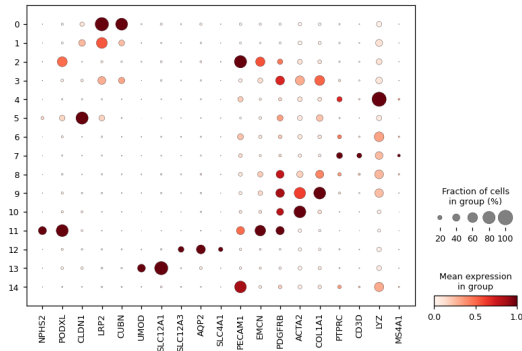
The 8 samples are separate tissue cores in one shared coordinate system. We split them with per-sample **bounding boxes** ('data/sample_bounding_boxes.csv'), derived from an independent, careful segmentation of the same slide: the cell barcodes differ, but the micron coordinates match, so the boxes transfer to the raw data. Each raw cell is assigned to the box that contains its centroid; the few cells in the gaps between cores are dropped.

Quality control



Transcripts and genes per cell (coverage / fragmented segments), control fraction (noise floor), and cell area (segmentation). Read the distributions and the per-sample summary before choosing thresholds; watch for a sample that stands apart for technical reasons.

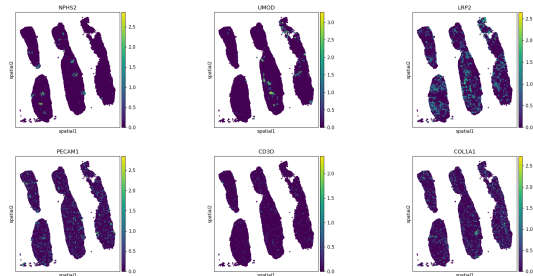
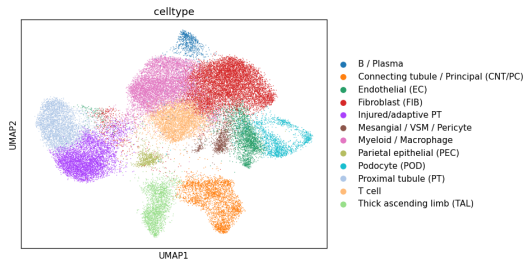
Annotation 1: clustering and markers



Cluster (Leiden) on the 480-gene panel (too few genes to select HVGs), then name clusters from **marker scores**: each cluster takes the lineage whose markers it expresses most. Transparent and reference-free, but coarse, the panel bounds what you can resolve.

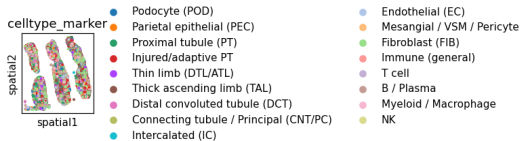
Check the dotplot: do clusters light up the canonical kidney markers (NPHS2/PODXL for podocytes, LRP2/CUBN for proximal tubule, UMOD for TAL, PECAM1 for endothelium)?

Cell types in space

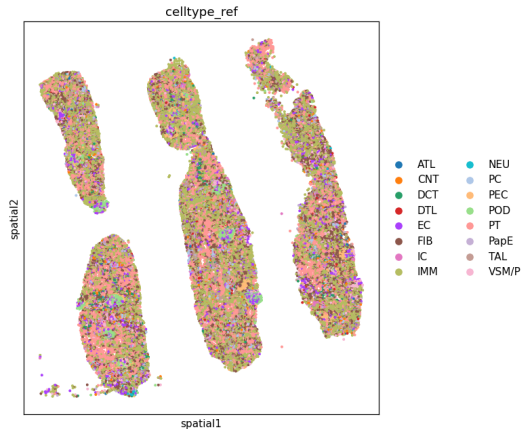


The UMAP groups cells transcriptionally; the tissue plot checks they form sensible structures. Tubule types should trace tubules, podocytes sit in glomerular tufts, immune cells are interstitial. A label that scatters where it should be structured is suspect.

Annotation 2: markers vs a reference



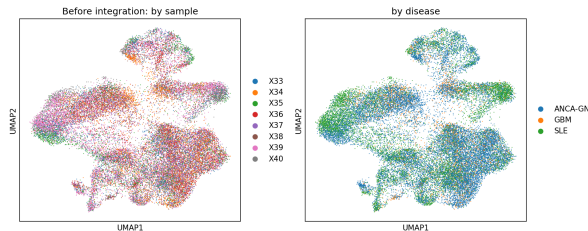
marker-based (reference-free)



reference-based (Lake/KPMP classifier)

A logistic-regression classifier trained on the Lake/KPMP snRNA-seq atlas ('subclass.l1', shared panel genes) transfers finer labels with a confidence. **Caveat:** a snRNA-seq vs imaging platform

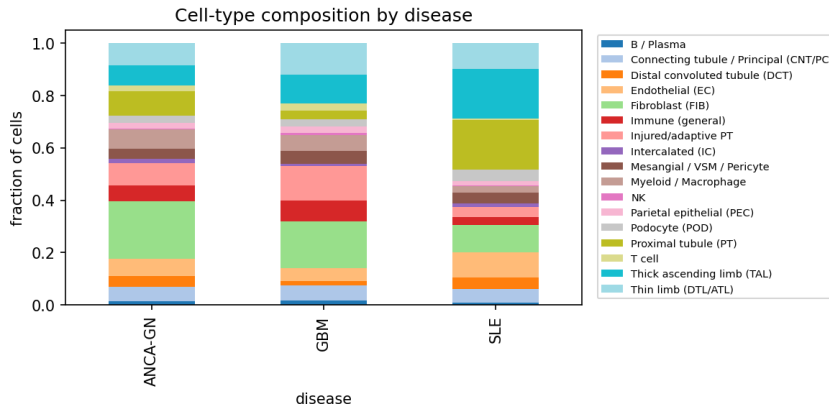
Integration: align samples before comparing



Before correction, cells separate partly by **sample/disease**: a batch effect mixed with real biology.

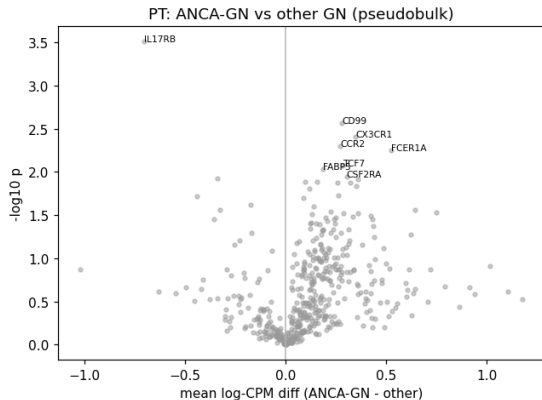
Harmony (CPU) aligns the embeddings so the same cell type overlaps across samples; scVI/scANVI are GPU alternatives. The risk is **over-correction** erasing real differences, so check that cell types and markers stay put.

Composition by disease



With cells aligned, compare cell-type fractions across groups. Read it as descriptive: proportions move with tissue sampling and the groups are small (5 ANCA-GN, 1 GBM, 2 SLE). The next step tests differences with the sample as the unit of replication.

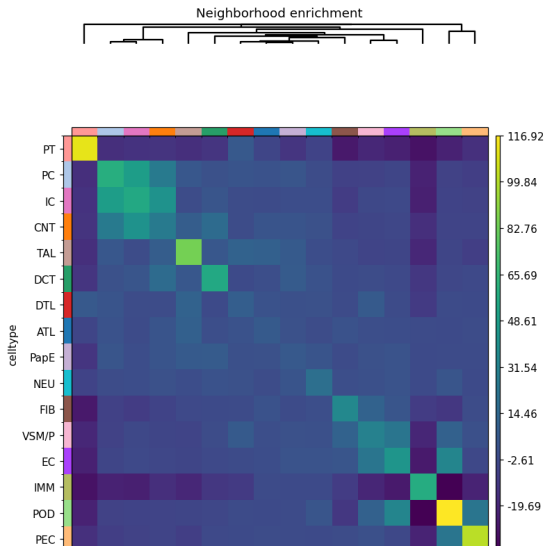
Differential expression: ANCA-GN vs other GN



Pseudobulk: sum counts per sample $\times$ cell type and test across *samples*. Per-cell DE would treat tens of thousands of cells from a few biopsies as independent (pseudoreplication) and vastly overstate significance.

With 5 vs 3 samples and no healthy control, treat hits as exploratory; a curated over-representation test then reads which programmes (inflammation, complement, fibrosis, endothelial activation) the changes touch.

Spatial: neighbourhood enrichment

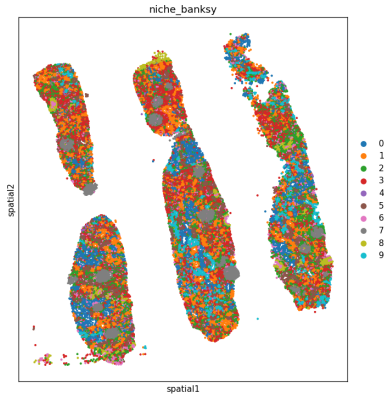
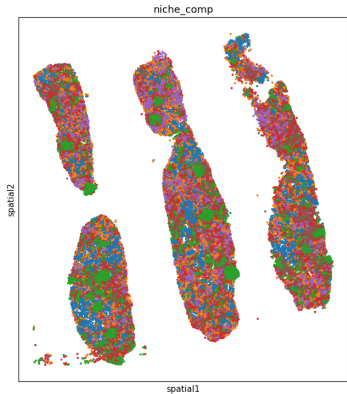


Permute cell-type labels on the spatial graph and compare adjacency to the null (z-score). The diagonal is self-aggregation; positive off-diagonal pairs co-locate (immune cells around injured tubules or vessels), negative pairs avoid each other.

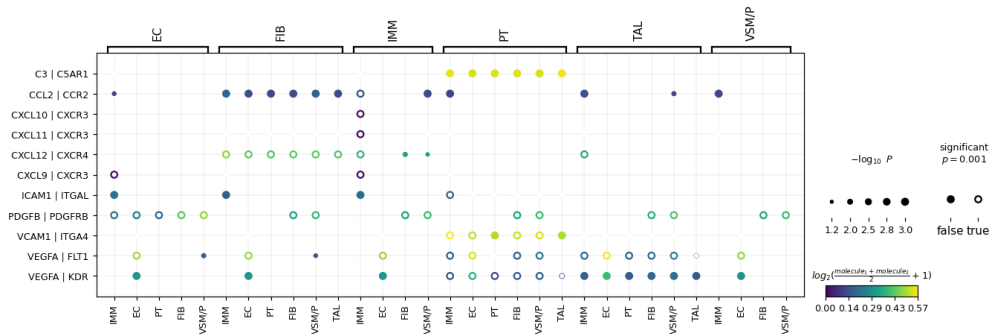
It is relative to this core's composition and the chosen graph, so compare like with like across samples.

Spatial: niches

A niche is a recurring local cell-type mixture. **Composition-based** clusters each cell's neighbourhood mix; **BANKSY-style** augments expression with the neighbour mean. They should agree on the major compartments (glomeruli, tubulo-interstitium, immune foci).



Spatial: ligand-receptor interactions

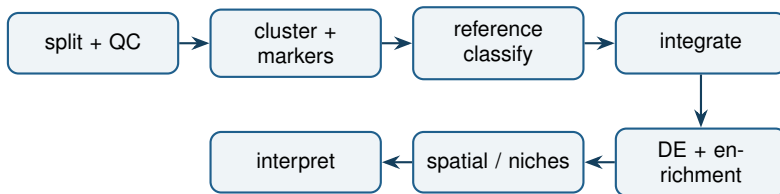


For each sender → receiver cell-type pair, test whether a ligand and its receptor are co-expressed more than chance (permutation). On a targeted panel only curated pairs are measurable (e.g. CXCL9/10/11 → CXCR3 recruiting T cells). Expression is a proxy for activity, and segmentation error can move a ligand to the wrong cell, so read hits as hypotheses.

Reading results critically

- **Replication is the sample, not the cell.** Per-cell tests overstate significance; use pseudobulk and composition-aware methods.
- **Small n, no control.** Five ANCA-GN vs three other-GN samples, no healthy tissue: contrasts are between GN types and are exploratory.
- **Segmentation and panel.** Misassigned transcripts corrupt typing and communication; absent genes are weak evidence.
- **Everything spatial is conditional** on the graph and the section. Compare across samples before drawing disease conclusions.

The workflow, and where to go next



The six notebooks follow this path on the real slide. Beyond the course: scANVI/scVI integration, whole-transcriptome enrichment (decoupler/gseapy + MSigDB), dedicated niche tools (BANKSY, CellCharter), and a healthy-control cohort for true case-control contrasts.